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EE1060: Differential Equations and Transform Techniques

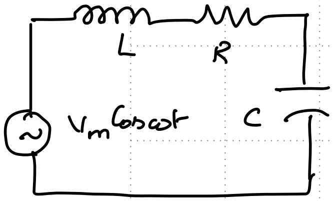
Lecture 15: Second-order nonhomogeneous ODEs

Topics :

1. General solution of Second-order nonhomogeneous ODE
2. Idea of Annihilation

General Solution of Second-order nonhomogeneous ODE and Notion of Linearity

Normal form : $y'' + P(x)y' + Q(x)y = g(x) \rightarrow$ linear & non homogeneous. $\hookrightarrow (1)$



$\Rightarrow L\dfrac{d^2 i}{dt^2} + R\dfrac{di}{dt} + \dfrac{i}{C} = -V_m\omega \sin\omega t \rightarrow$ Linear and non homogeneous.

(input $\rightarrow V_m \sin\omega t$)

output $\rightarrow$ 'i'

Any system whose i/p - o/p relation is characterized by linear DE is a linear system.

if $y(x)$ is a solution when $g(x) = g_1(x)$

then $\alpha y(x)$ " " " " $g(x) = \alpha g_1(x)$

if y_1 & y_2 are solutions of (1), then $\alpha y_1 + \beta y_2$ is a sol to $\alpha g_1 + \beta g_2$.

General Solution :- if $\{y_1, y_2\} \leftarrow$ fundamental set of sol for $L[y] = 0$ and y_p is any sol to $L[y] = g(x)$ then every sol for $L[y] = g(x)$ can be expressed as

$$y(x) = \underbrace{\sum_{i=1}^{n} c_i y_i(x)}_{\text{Complementary solution.}} + y_p(x)$$

Proof :- Let $y(x)$ be a sol. to $L[y] = g(x)$

$L[y_p] = g(x)$

$\Rightarrow L[y] - L[y_p] = 0$

$\Downarrow$ Linearity

$\Rightarrow L[y - y_p] = 0. \Rightarrow y - y_p = \sum_{i=1}^{n} c_i y_i$

Method of Undetermined Coefficients - Annihilator (Definition and Annihilator of x^n)

only for DE with Const Coefficients.

$$D = \frac{d}{dx}$$

2nd order Normal form: $y'' + p(x)y' + q(x)y = g(x)$

$$\Rightarrow (D^2 + p(x)D + q(x))[y] = g(x)$$

for the case of Const Coefficient ODE $\Rightarrow \underbrace{(D^2 + pD + q)}_{\text{Characteristic Eqn.}}[y] = g(x)$

An Annihilator $L(D)$ is a function of 'D' such that

$$L(D)[g(x)] = 0.$$

$\hookrightarrow$ if $\underbrace{g(x), g'(x), \ldots g^{n}(x)}_{\text{are Linearly dependent.}}$

don't Know.

if $g(x) = x^n$, $L(D) = D^{n+1}$

if $g(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$, $L(D) = D^{n+1}$

Annihilator of e^{ax} and $x^n e^{ax}$

$g(x) = e^{ax}$:

$$g'(x) = D[g(x)] = ae^{ax}.$$

$$-ag(x) = \underline{}$$

$$(0)$$

$$L(D) = (D-a)$$

$g(x) = xe^{ax}$

$L(D) = (D-a)^2$ $-2a.$

$$g(x) = xe^{ax}.$$

$$g'(x) = e^{ax} + axe^{ax}.$$

$$g''(x) = 2ae^{ax} + a^2 x e^{ax}$$

$L(D)[g(x)] = 0$ for.

$$g''(x) - 2a g'(x) + a^2 g(x) = 0$$

$$= (D^2 - 2aD + a^2) g(x) = 0$$

$$\underbrace{(D-a)^2}\, g(x) = 0.$$

$g(x) = x^n e^{ax}$

$L(D) = (D-a)^{n+1}$

Annihilator of $\sin mx$ and $\cos mx$

$g(x) = \cos mx.$

$D^2 + m^2$

$g(x) = \sin mx$

$g(x) = \cos mx$

$g'(x) = -m \sin mx$

$g''(x) = -m^2 \cos mx$

$L(D) = (D^2 + m^2)$

$g(x) = e^{ax} \cos mx$

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$(D-a) \quad (D^2+m^2)$

$g(x) = e^{ax} \cos mx$

$g'(x) = a e^{ax} \cos mx - m e^{ax} \sin mx$

$g''(x) = a^2 e^{ax} \cos mx - 2am\, e^{ax} \sin mx - m^2 e^{ax} \cos mx.$

$g'''(x) = a^3 e^{ax} \cos mx - 3a^2 m\, e^{ax} \sin mx - 3am^2 e^{ax} \cos mx + m^3 e^{ax} \sin mx.$

$L(D) = \quad g''(x) - 2a\, g'(x) + (m^2 + a^2)\, g(x) = 0 \; (\checkmark)$

$L(D) = \quad g'''(x)$